

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA1402

COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO4: *Examine intermediate code generation techniques to enhance code optimization (BL4)*

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

CLASS TEST

Code of Conduct:

*I __Monika.R(192324281)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:



Q. No	Understanding of Concepts (25)	Explanation (25)	Application (20)	Organization (15)	Accuracy(10)	Total Score (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/ 30	/ 20	/ 15	/ 10	/ 100

CO4-AT1 - CLASS TEST

①. construct three Address code

if $((a < b) \text{ and } ((c < d) \text{ or } (a > d)))$ then

$z = x + y * z$

else

$z = z + 1$

Given:

$(a < b) \text{ AND } ((c < d) \text{ OR } (a > d))$

$a < b$

$c < d$

$a > d$

The structure is $(a < b) \text{ AND } [(c < d) \text{ OR } (a > d)]$.

Generate TAC for the condition:

L1:

if $a < b$ goto L2

goto L5

L2:

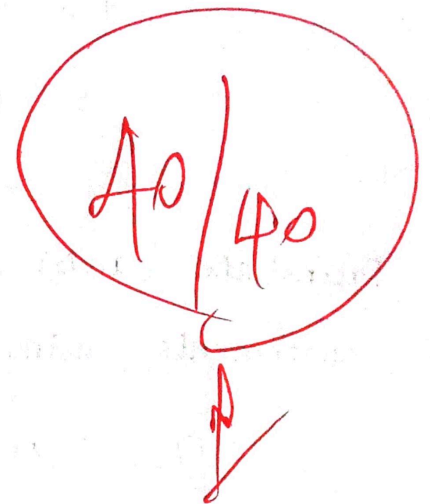
if $c < d$ goto L4

goto L3

L3:

if $a > d$ goto L5

goto L5



Generate TAC for the THEN part

L4:

$$t_1 = y * z$$

$$t_2 = x + t_1$$

$$z = t_2$$

goto L6

Generate TAC for the ELSE part

L5:

$$t_3 = z + 1$$

$$z = t_3$$

- ②. Translate $(a < b)$ and $(c < d)$ or $(e < f)$ into three address statements using backpatching.

$[(a < b) \text{ AND } (c < d)] \text{ OR } (e < f)$

Generate incomplete jumps:

1. if $a < b$ goto _
2. goto _
3. if $c < d$ goto _
4. goto _
5. if $e < f$ goto _
6. goto _

The underscore represents a target address that is not known yet.

Handle AND:

For $(a < b)$ AND $(c < d)$, if $a < b$ is true, continue to the second condition.

1. if $a < b$ goto 3

Handle OR:

If the first AND expression $\rightarrow$ false

OR operand $e < f$ must be evaluated.

Backpatch statement 2 to statement 5

Backpatch statement 4 to statement 5

Final backpatched code:

1. if $a < b$ goto 3

2. goto 5

3. if $c < d$ goto 7

4. goto 5

5. if $e < f$ goto 7

6. goto 8

7. TRUE

8. FALSE.

③. construct three address code using the following production for Boolean expression $(a > b)$ or $(c < d)$ or $(e < f)$

$B \rightarrow B_1 \parallel B_2$

$B \rightarrow B_1 \&\& B_2$

$B \rightarrow !B_1$

$B \rightarrow E_1 \text{ xelop } E_2$

$B \rightarrow \text{true}$

$B \rightarrow \text{false}$

$[(a > b) \text{ OR } (c < d)] \text{ OR } (e < f)$

Generate TAC :

L1 :
if $a > b$ goto LTRUE
goto L2

L2 :
if $c < d$ goto LTRUE
goto L3

L3 :
if $e < f$ goto LTRUE
goto LFALSE

LTRUE : // Boolean expression is true

LFALSE : // Boolean expression is false.

4.

Generate TAC for the following program fragment -

while ($A < C$ and $B > D$) do

if $A = 1$ then $C = C + 1$

else while $A \leq D$ do

$A = A + B$

The outer condition is $A < C$ AND $B > D$

L1 :

if $A < C$ goto L2

goto LEND

L2 :

if $B > D$ goto L3

goto LEND

if condition:

L3 :

if $A == 1$ goto L4

goto L5

THEN block:

L4 :

$t_1 = C + 1$

$C = t_1$

goto L1

ELSE block:

L5 :

$t_2 = A + B$

$A = t_2$

goto L5

inner while :

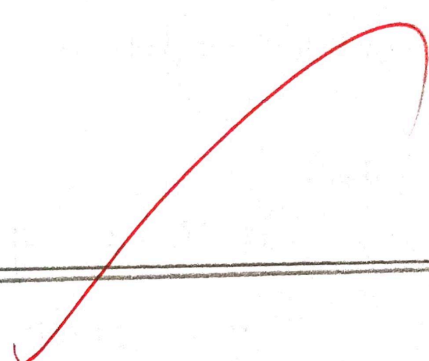
L₆ :

$$t_2 = A + B$$

$$A = t_2$$

goto L₅

LEND :



RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	15%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps